

Splunk Failed Login & Brute-Force Detection Lab

SOC-style authentication log investigation using Splunk Enterprise, SPL, MITRE ATT&CK mapping, and incident reporting.

Analyst: Vedant Rawal

Role target: SOC Analyst Intern / Cybersecurity Analyst Intern

Dataset: failed_login_lab

Tool: Splunk Enterprise

Severity of key finding: Critical

Primary finding: Successful admin VPN login after 45 previous failed attempts from the same source IP.

Executive Summary

A Splunk-based authentication log investigation identified a critical suspicious login pattern. Source IP 45.77.32.11 generated 45 failed login attempts, targeted 9 unique users, and later produced a successful login for user admin on vpn-gateway.

This pattern may indicate brute-force success, password spraying, credential stuffing, or use of valid compromised credentials.

Key Metrics

Total authentication events: 195

Failed login events: 96

Successful login events: 99

Unique source IPs: 7

Unique users: 20

Top suspicious source IP: 45.77.32.11

Highest failed login count: 45

Successful login after failures: Yes - admin on vpn-gateway

Severity: Critical

SPL Detection Queries

1) Summary Statistics:

index=failed_login_lab | stats count as total_events, count(eval(action="failure")) as failed_logins, count(eval(action="success")) as successful_logins, dc(src_ip) as unique_source_ips, dc(user) as unique_users

2) Failed Logins by Source IP:

index=failed_login_lab action=failure | stats count by src_ip | sort - count

3) Failed Logins by User:

index=failed_login_lab action=failure | stats count by user | sort - count

4) Suspicious IPs Targeting Multiple Users:

index=failed_login_lab action=failure | stats count dc(user) as unique_users values(user) as targeted_users by src_ip | where count > 20 OR unique_users > 5 | sort - count

5) Successful Login After Failures:

index=failed_login_lab | sort 0 _time | streamstats count(eval(action="failure")) as previous_failures by src_ip | where action="success" AND previous_failures>=20 | table _time src_ip user dest_host app action previous_failures reason severity country

Findings

Finding 1: Source IP 45.77.32.11 generated 45 failed login attempts and later succeeded.

Finding 2: Source IP 185.220.101.23 generated 30 failed login attempts.

Finding 3: 45.77.32.11 targeted 9 accounts: admin, administrator, backup, devops, oracle, postgres, root, test, vedant.

Finding 4: Successful login details - time: 2026-06-26 11:50:00.450, source IP: 45.77.32.11, user: admin, destination: vpn-gateway, app: vpn, previous failures: 45, reason: valid_password_after_failures, severity: critical, country: Netherlands.

MITRE ATT&CK Mapping

T1110 - Brute Force: Multiple failed login attempts from the same source IPs.

T1078 - Valid Accounts: A successful login occurred after repeated failures, suggesting possible credential compromise.

T1133 - External Remote Services: VPN gateway authentication was involved.

T1589 - Gather Victim Identity Information: Multiple usernames were targeted by suspicious

source IPs.

Business Impact

Possible unauthorized VPN access, account takeover, exposure of internal services, lateral movement risk, and compliance/reporting impact.

Privileged usernames such as admin, administrator, root, and itadmin increase risk because compromise of these accounts can expand attacker access.

Recommended Response Actions

1. Confirm whether the admin VPN login was authorized.
2. Force password reset for affected accounts and review privileged accounts.
3. Block or rate-limit suspicious IPs at firewall, VPN, or identity provider level.
4. Enable MFA for VPN and all privileged accounts.
5. Create a SIEM alert for repeated failures followed by success.
6. Review VPN logs for post-authentication activity.
7. Build baseline detection for abnormal login behavior and geo-anomalies.

Conclusion

This lab demonstrates a complete SOC-style workflow: importing authentication logs, writing SPL queries, detecting brute-force behavior, identifying successful login after multiple failed attempts, mapping behavior to MITRE ATT&CK, and documenting impact and remediation.

Portfolio positioning: Beginner SOC analyst candidate with hands-on Splunk, SIEM investigation, authentication log analysis, MITRE ATT&CK mapping, and incident reporting practice.